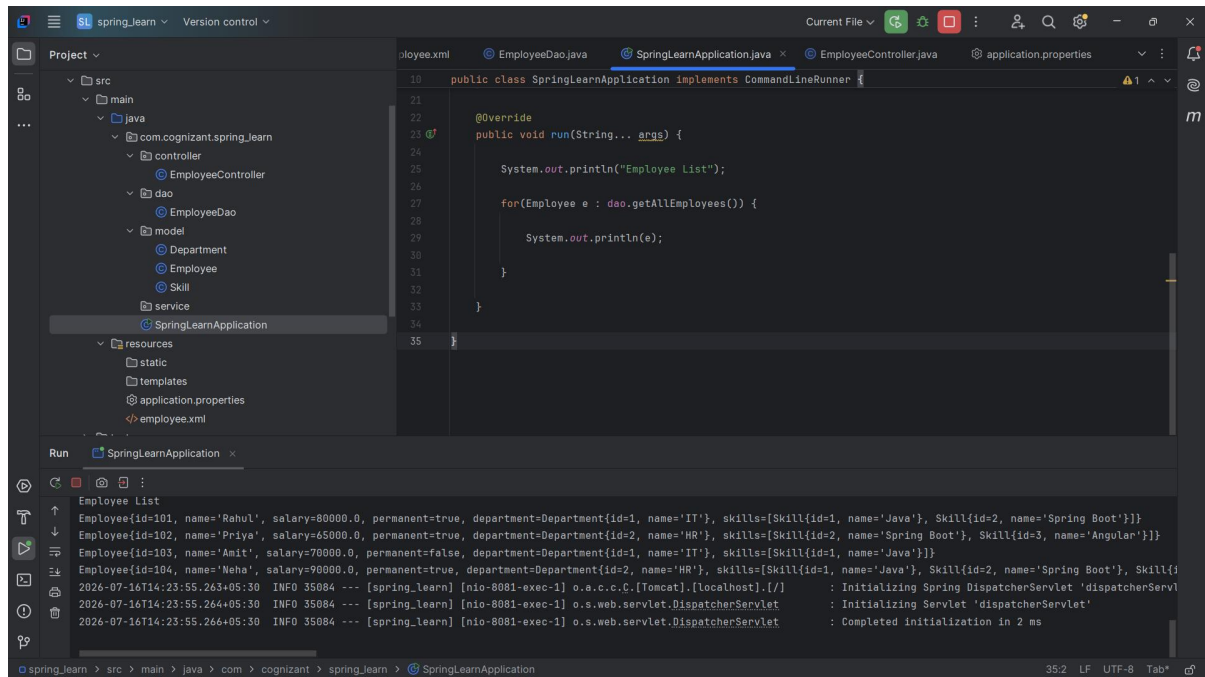


Spring REST using Spring Boot 3

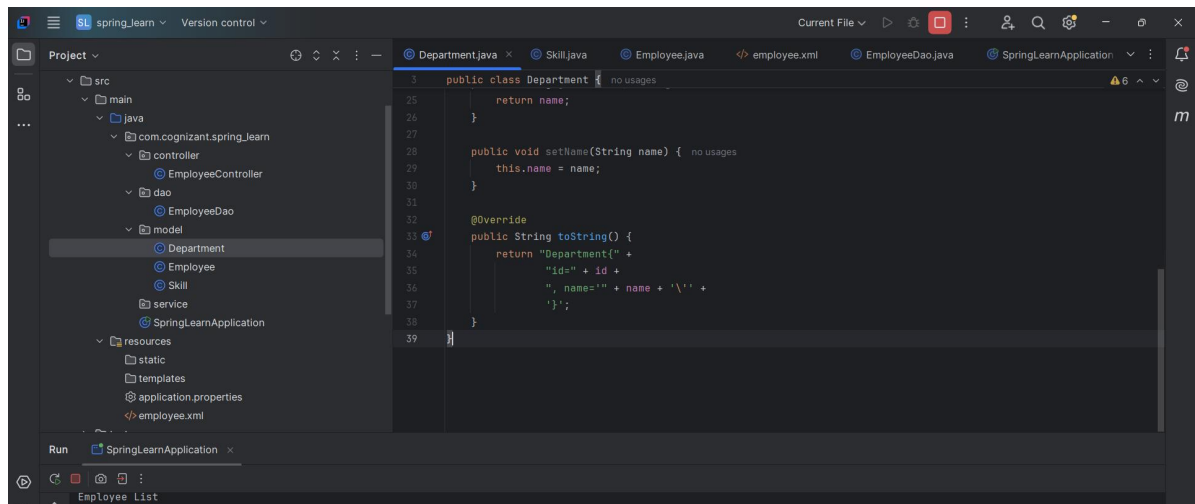
3. spring-rest-handson

1. Create static employee list data using spring xml configuration.

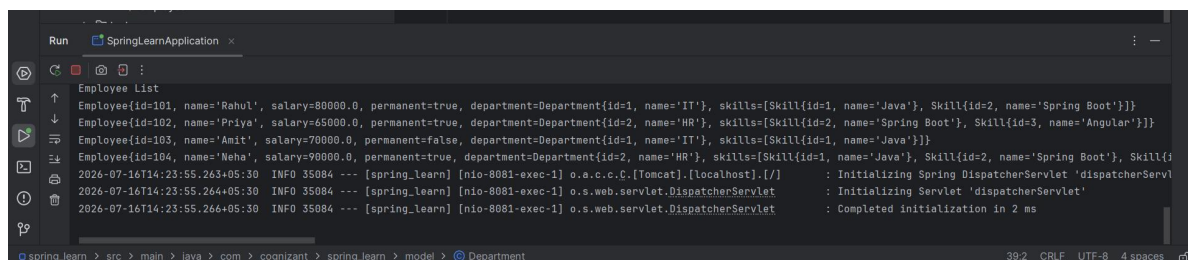


```
10 public class SpringLearnApplication implements CommandLineRunner {
21
22
23     @Override
24     public void run(String... args) {
25
26         System.out.println("Employee List");
27
28         for(Employee e : dao.getAllEmployees()) {
29
30             System.out.println(e);
31
32         }
33     }
34
35 }
```

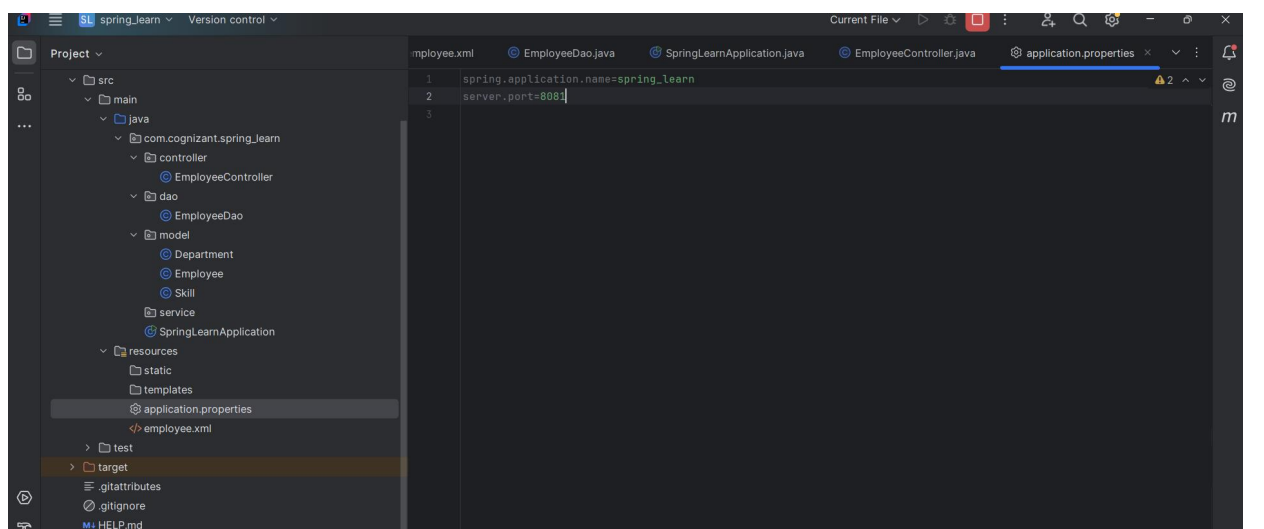
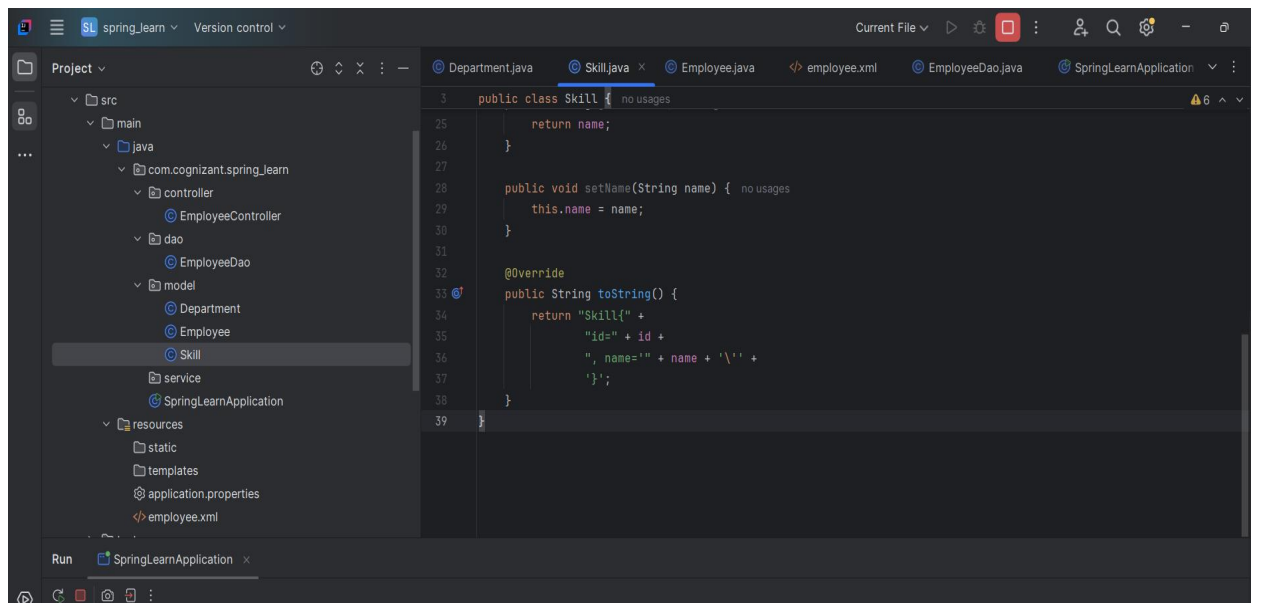
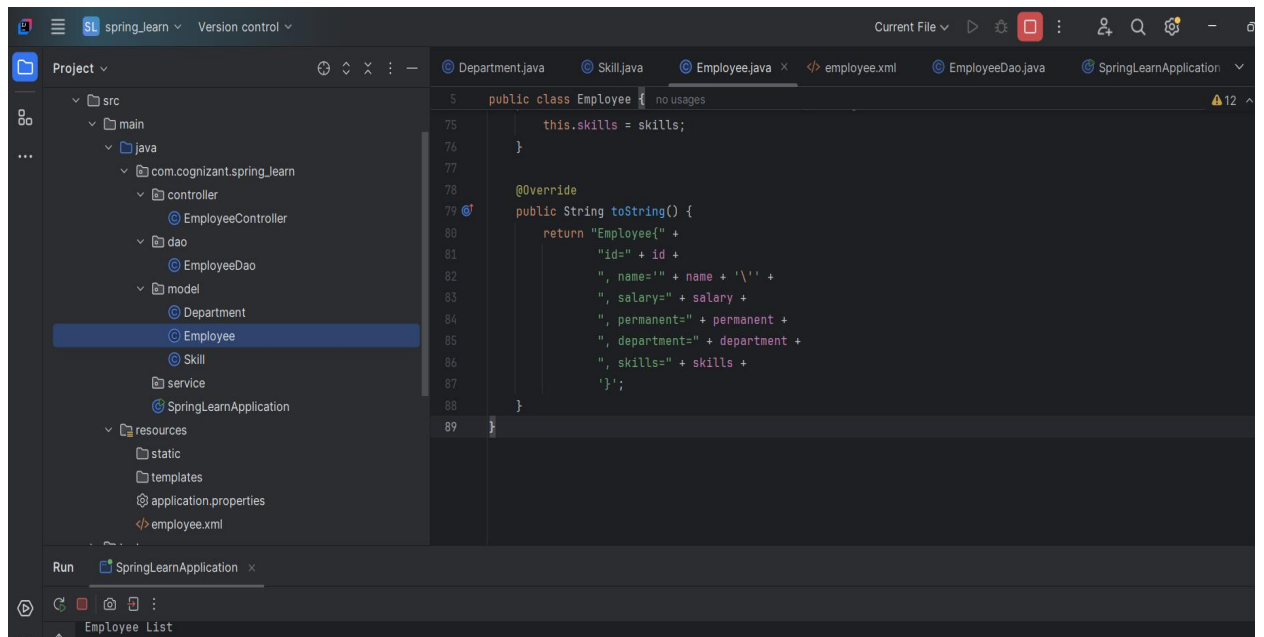
```
Employee List
Employee{id=101, name='Rahul', salary=80000.0, permanent=true, department=Department{id=1, name='IT'}, skills=[Skill{id=1, name='Java'}, Skill{id=2, name='Spring Boot'}]}
Employee{id=102, name='Priya', salary=65000.0, permanent=true, department=Department{id=2, name='HR'}, skills=[Skill{id=2, name='Spring Boot'}, Skill{id=3, name='Angular'}]}
Employee{id=103, name='Amit', salary=70000.0, permanent=false, department=Department{id=1, name='IT'}, skills=[Skill{id=1, name='Java'}]}
Employee{id=104, name='Neha', salary=90000.0, permanent=true, department=Department{id=2, name='HR'}, skills=[Skill{id=1, name='Java'}, Skill{id=2, name='Spring Boot'}, Skill{id=3, name='Angular'}]}
2026-07-16T14:23:55.263+05:30 INFO 35084 --- [spring_learn] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-16T14:23:55.264+05:30 INFO 35084 --- [spring_learn] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-16T14:23:55.266+05:30 INFO 35084 --- [spring_learn] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
```

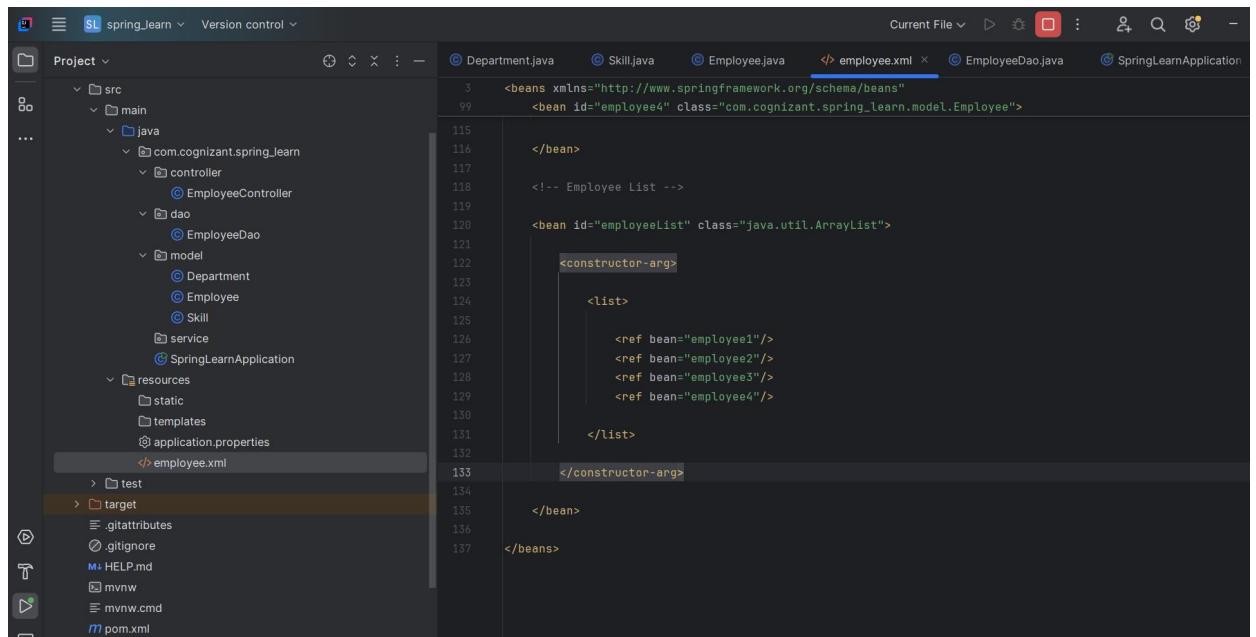


```
1 public class Department {
2
3     return name;
4
5 }
6
7 public void setName(String name) {
8     this.name = name;
9 }
10
11 @Override
12 public String toString() {
13     return "Department{" +
14         "id=" + id +
15         ", name='" + name + '\'' +
16     "}"
17 }
18
19 }
```

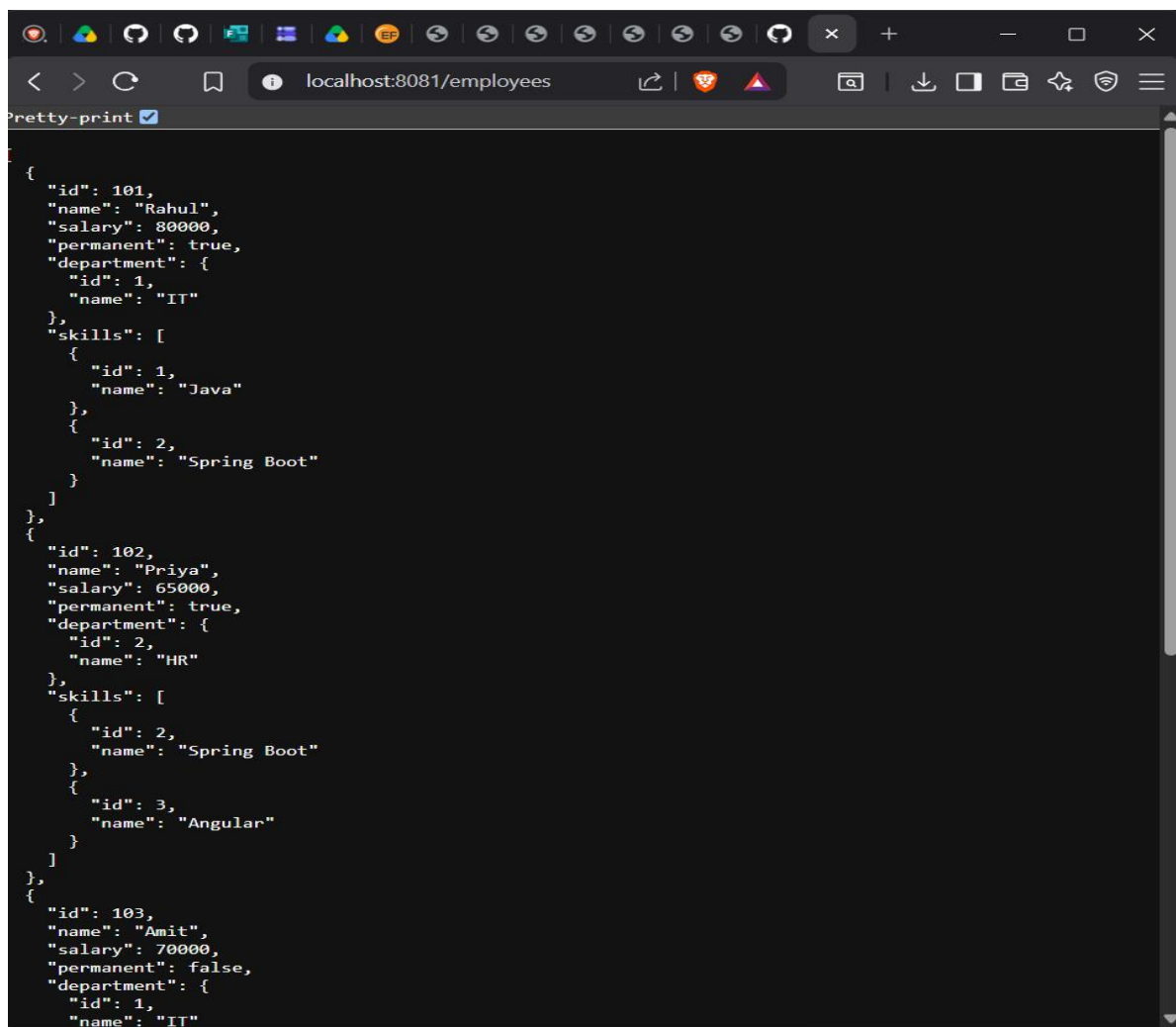


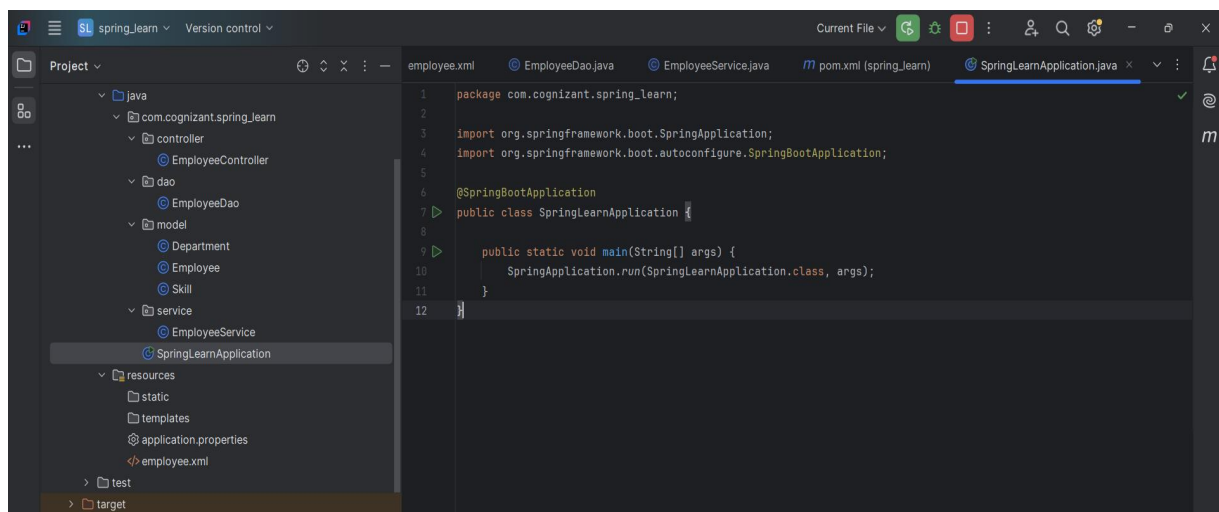
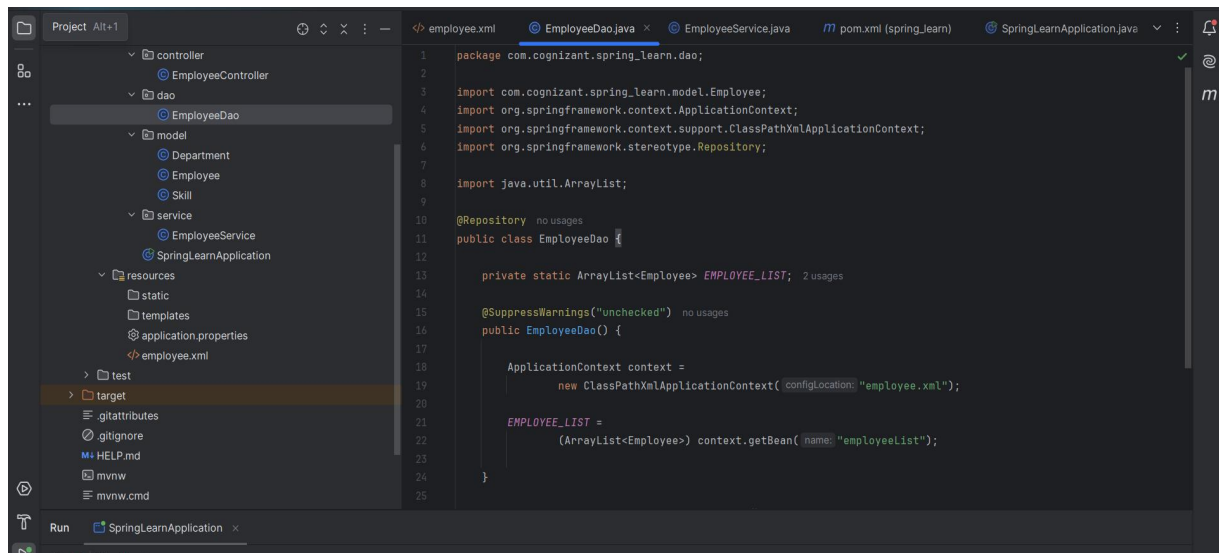
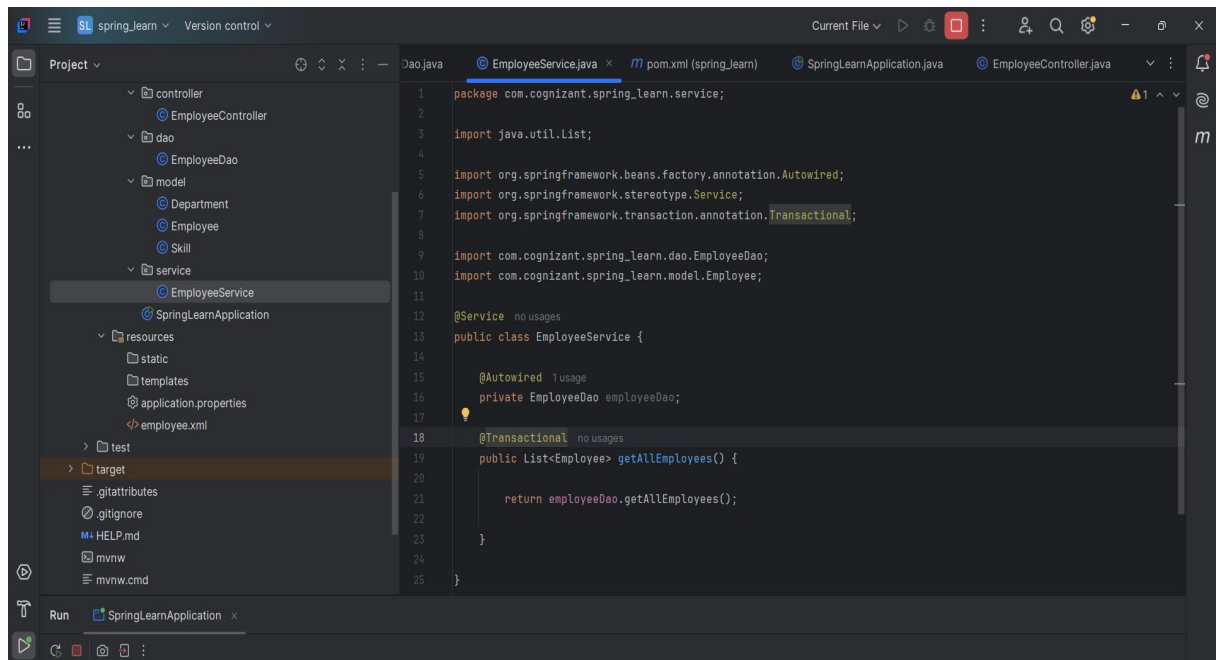
```
Employee List
Employee{id=101, name='Rahul', salary=80000.0, permanent=true, department=Department{id=1, name='IT'}, skills=[Skill{id=1, name='Java'}, Skill{id=2, name='Spring Boot'}]}
Employee{id=102, name='Priya', salary=65000.0, permanent=true, department=Department{id=2, name='HR'}, skills=[Skill{id=2, name='Spring Boot'}, Skill{id=3, name='Angular'}]}
Employee{id=103, name='Amit', salary=70000.0, permanent=false, department=Department{id=1, name='IT'}, skills=[Skill{id=1, name='Java'}]}
Employee{id=104, name='Neha', salary=90000.0, permanent=true, department=Department{id=2, name='HR'}, skills=[Skill{id=1, name='Java'}, Skill{id=2, name='Spring Boot'}, Skill{id=3, name='Angular'}]}
2026-07-16T14:23:55.263+05:30 INFO 35084 --- [spring_learn] [nio-8081-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-16T14:23:55.264+05:30 INFO 35084 --- [spring_learn] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-16T14:23:55.266+05:30 INFO 35084 --- [spring_learn] [nio-8081-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 2 ms
```



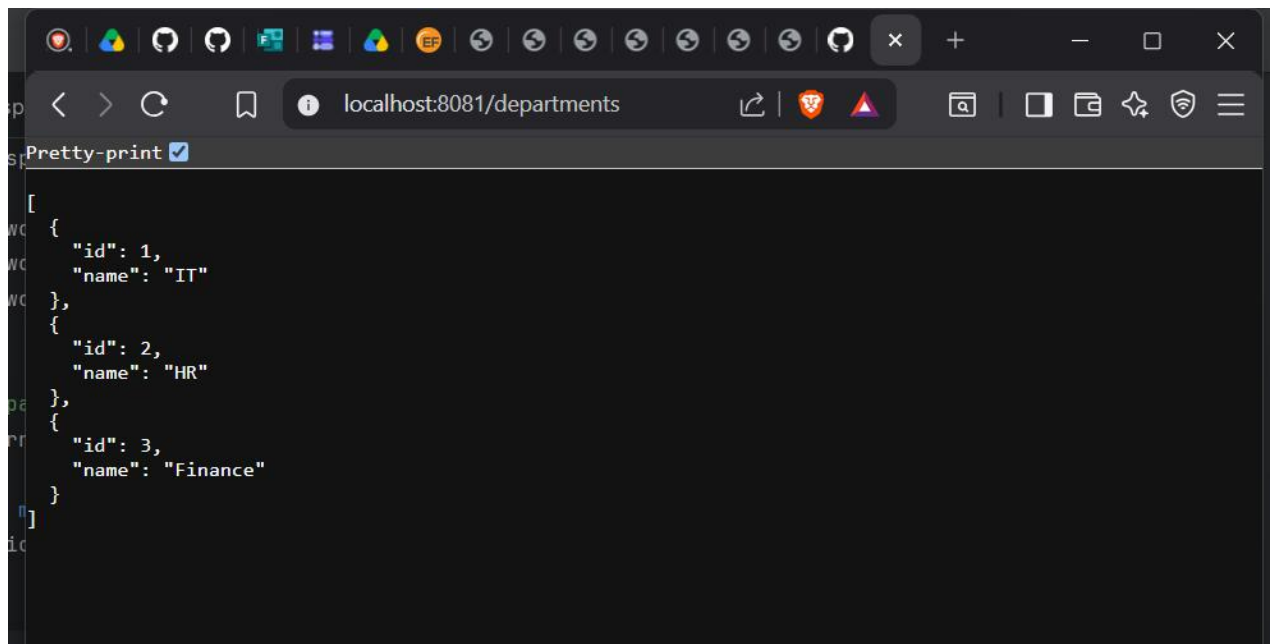


2.Create REST service to gets all employees.



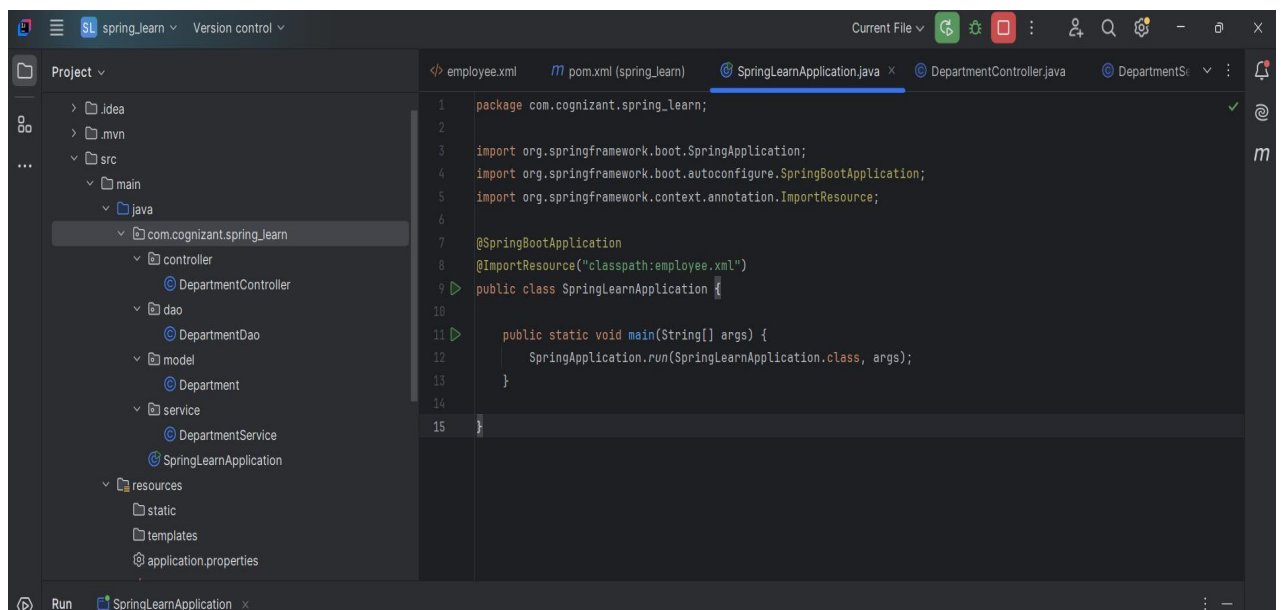


3. Create REST service for department



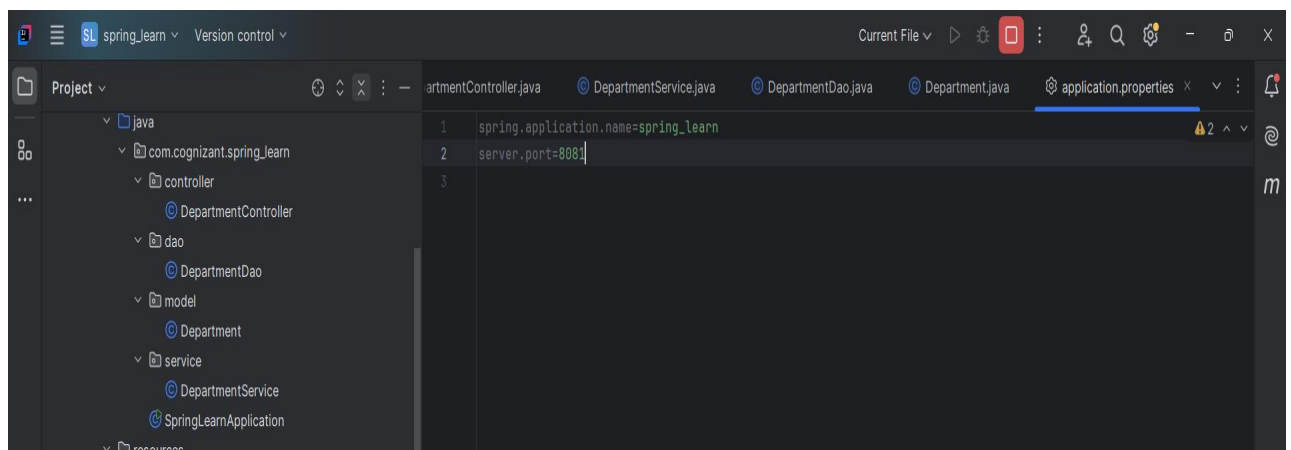
A screenshot of a web browser window displaying a REST API response. The address bar shows the URL `localhost:8081/departments`. The response is a JSON array of department objects, which has been formatted using a "Pretty-print" tool. The JSON data is as follows:

```
[
  {
    "id": 1,
    "name": "IT"
  },
  {
    "id": 2,
    "name": "HR"
  },
  {
    "id": 3,
    "name": "Finance"
  }
]
```



A screenshot of an IDE (IntelliJ IDEA) showing the `SpringLearnApplication.java` file. The project structure on the left includes `com.cognizant.spring_learn` with sub-packages `controller`, `dao`, `model`, `service`, and `resources`. The main class `SpringLearnApplication` is highlighted. The code in the editor is as follows:

```
1 package com.cognizant.spring_learn;
2
3 import org.springframework.boot.SpringApplication;
4 import org.springframework.boot.autoconfigure.SpringBootApplication;
5 import org.springframework.context.annotation.ImportResource;
6
7 @SpringBootApplication
8 @ImportResource("classpath:employee.xml")
9 public class SpringLearnApplication {
10
11     public static void main(String[] args) {
12         SpringApplication.run(SpringLearnApplication.class, args);
13     }
14
15 }
```



A screenshot of an IDE (IntelliJ IDEA) showing the `application.properties` file. The project structure on the left is the same as in the previous screenshot. The code in the editor is as follows:

```
1 spring.application.name=spring_learn
2 server.port=8081
3
```